Multiplying and Dividing Algebraic Fractions

Y10

To multiply algebraic fractions, multiply the numerator by themselves and the denominator by themselves.

Examples.

$\frac{3a}{5} \times \frac{c}{2}$

$= \frac{3ac}{10}$

$\frac{6p}{5} \times \frac{10c}{2pq}$

$= \frac{{}^{3}\cancel{6}\cancel{p}^{1}}{{}_{1}\cancel{5}} \times \frac{{}^{2}\cancel{10}c}{{}_{1}\cancel{2}\cancel{p}^{1}q}$

$= \frac{6c}{q}$

$\frac{2v}{(a+b)} \times \frac{2a+2b}{4uv}$

$= \frac{{}^{1}\cancel{2}\cancel{v}^{1}}{{}_{1}\cancel{(a+b)}} \times \frac{{}^{1}\cancel{2}\cancel{(a+b)}^{1}}{{}_{1}\cancel{{}^{2}}\cancel{4}u\cancel{v}}$

$= \frac{1}{u}$

A. Simplify the following.

1). $\frac{2a}{5} \times \frac{b}{3}$ 2). $\frac{2u}{3} \times \frac{v}{4}$ 3). $\frac{3x}{2y} \times \frac{y}{6}$ 4). $\frac{8a}{3} \times \frac{2b}{12}$ 5). $\frac{12x}{5} \times \frac{y}{2x}$

6). $\frac{3p}{5} \times \frac{10}{2pq}$ 7). $\frac{8c}{4} \times \frac{9c}{3cd}$ 8). $\frac{5v}{6} \times \frac{18u^2}{3uv}$ 9). $\frac{6p^2}{q} \times \frac{10q}{20p}$ 10). $\frac{5b}{2} \times \frac{14a}{7a^2b}$

11). $\frac{2v}{6(u+v)} \times \frac{3u+3v}{4uv}$ 12). $\frac{6r}{3s+3t} \times \frac{5s+5t}{2rt}$ 13). $\frac{2}{4u+2} \times \frac{(2u+1)}{u^2}$

14). $\frac{15a^2}{2a+b} \times \frac{4a+2b}{10a}$ 15). $\frac{2wx}{u^2-v^2} \times \frac{2u+2v}{4w}$ 16). $\frac{2a}{a-b} \times \frac{a^2-b^2}{6ab}$

To divide algebraic fractions multiply by the reciprocal of the divisor.

Examples

$\frac{2a}{7} \div \frac{3}{c}$

$\frac{2a}{7} \times \frac{c}{3}$

$= \frac{2ac}{21}$

$\frac{3p}{4} \div \frac{2pq}{12c}$

$\frac{3p}{4} \times \frac{12c}{2pq}$

$= \frac{3\cancel{p}^{1}}{{}_{1}\cancel{4}} \times \frac{{}^{3}\cancel{12}c}{2\cancel{p}^{1}q}$

$= \frac{9c}{2q}$

$\frac{2v}{3(a-b)} \div \frac{8v}{6a-6b}$

$\frac{2v}{3(a-b)} \times \frac{6a-6b}{8v}$

$= \frac{{}^{1}\cancel{2}\cancel{v}^{1}}{{}_{1}\cancel{3}\cancel{(a-b)}^{1}} \times \frac{{}^{1}\cancel{{}^{2}}\cancel{6}\cancel{(a-b)}^{1}}{{}_{2}\cancel{4}\cancel{8}\cancel{v}^{1}}$

$= \frac{1}{2}$

B. Simplify the following.

1). $\frac{2a}{7} \div \frac{a}{14}$ 2). $\frac{2x}{3} \div \frac{4}{y}$ 3). $\frac{2}{a} \div \frac{6}{a^2}$ 4). $\frac{uv}{5} \div \frac{v}{15}$ 5). $\frac{2a^2}{8} \div \frac{ab}{4}$

6). $\frac{3p}{4} \div \frac{6pq}{12q}$ 7). $\frac{2s}{t} \div \frac{8s^2}{5t}$ 8). $\frac{2p}{5} \div \frac{p^2q}{10q}$ 9). $\frac{bd}{4c} \div \frac{2bd^2}{12c^2}$ 10). $\frac{9g^2}{8f} \div \frac{6gh}{12f}$

11). $\frac{8a}{(a-b)} \div \frac{6a}{3a-3b}$ 12). $\frac{3x}{4(p+2)} \div \frac{6x}{3p+6}$ 13). $\frac{2p-4}{3p-9} \div \frac{7p-14}{5p-15}$

14). $\frac{6x}{x-y} \div \frac{12x^2}{x^2-y^2}$ 15). $\frac{2p^2}{p^2-q^2} \div \frac{12pq}{6p+6q}$ 16). $\frac{5}{2x+1} \div \frac{2x-1}{4x^2-1}$